

The margin–sensitivity coupling: resolved results, the one controlling quantity, and the remaining open path

Working note (theory worked out by hand; agents used only for literature/refutation checks)

June 28, 2026

Abstract

This note isolates the “self-limiting coupling” theory behind the paper’s margin–sensitivity discussion. Earlier drafts stated four conjectures (A–D); this version records what is now settled. The single controlling quantity is the *spherical log-margin gradient* $\|\nabla_{\mathbb{S}} \log m(u)\|$, equivalently $\tan \angle(\nabla M(x), x)$: the upper envelope on $\|\nabla M\|$ holds iff this is bounded, and the same quantity controls the certificate’s looseness. We give the exact identity (§4), a sharp linear case, a conditional bound under global max-margin, an explicit counterexample showing no universal constant exists, the corrected lockstep statement (§5), the corrected local capacity statement (§6), and the bracket-condition-number result (§7). The one genuinely open path is an implicit-bias bound on the spherical log-gradient at the gradient-descent solution (§8). Everything labelled Proposition/Theorem here is proved; the open item is stated as such.

1 Setup

Let $f : \mathbb{R}^d \rightarrow \mathbb{R}^k$ be a classifier with logits $f_1, \dots, f_k$ and, for a labelled point (x, y) , signed margin $M(x) = f_y(x) - \max_{j \neq y} f_j(x)$, so $M > 0$ iff f is correct. Write $\eta = \mathbb{E}[M]$ over correct points and $L = \mathbb{E}\|\nabla M\|$ for the mean input-gradient norm (the local sensitivity; norm dual to the threat model). The first-order certificate is $r_2 \geq M(x)/\|\nabla M(x)\|_2$ (Hein–Andriushchenko 2017; Tsuzuku et al. 2018), so η/L is a robustness surrogate. The empirical finding is that η/L *diagnoses* robustness across architectures yet *training* it does not buy robustness; the question is what, precisely, is true behind that.

2 The radial envelope (Euler)

Proposition 1 (Homogeneity lower envelope). *Let f be bias-free with degree-one positively homogeneous activations (ReLU, leaky-ReLU, absolute value, maxout). Then M is degree-one positively homogeneous, $\langle \nabla M(x), x \rangle = M(x)$ (Euler) holds a.e., and hence*

$$\|\nabla M(x)\|_2 \geq \frac{M(x)}{\|x\|_2}, \quad \|\nabla M(x)\|_1 \geq \frac{M(x)}{\|x\|_\infty}, \quad \eta/L \leq \sup_i \|x_i\|_2.$$

This is exact and standard (Euler’s identity plus Cauchy–Schwarz / Hölder). The content is only the reading that the margin gradient has an architecture-imposed *radial* floor. It controls the radial component of the gradient and nothing else, which is exactly the limitation made precise in §4.

3 Why the “impossibility” reading of the coupling is wrong

An earlier draft read Proposition 1 as forbidding the training of η/L . It does not.

- (W1) *Wrong direction.* To raise $\eta/L = M/\|\nabla M\|$ one wants $\|\nabla M\|$ small. Proposition 1 is a *lower* bound on $\|\nabla M\|$, i.e. an *upper* cap $\eta/L \leq \|x\|$; an upper cap above the operating value cannot forbid increasing the ratio.
- (W2) *Loose at the operating point.* Empirically $\|\nabla M\|_2 \approx 26 M/\|x\|_2$, an order of magnitude of slack.
- (W3) *Hypotheses fail.* Euler needs bias-free, exactly degree-1 homogeneous; the trained models carry biases and batch normalization.
- (W4) *Static vs. dynamic.* The empirical obstruction is a property of the training path, not of a static inequality.

What survives unconditionally is the scale degeneracy.

Lemma 1 (Scale degeneracy of the ratio). *$M(x)/\|\nabla M(x)\|$ is invariant under $f \mapsto cf$ ($c > 0$). Hence “maximize η/L ” (equivalently penalize $\|\nabla M\|/M$) is blind to absolute margin and admits the constant classifier $\nabla M \equiv 0$ as a global optimizer; direct ratio maximization is ill-posed. (Tsuzuku et al. 2018, App. G.2.)*

4 The exact spherical envelope (resolves A)

The radial floor controls only the radial gradient. Writing the gradient in polar form isolates the missing piece exactly.

Proposition 2 (Radial–tangential split of the margin gradient). *Let $M : \mathbb{R}^d \setminus \{0\} \rightarrow \mathbb{R}$ be C^1 and positively 1-homogeneous. For $x \neq 0$ set $r = \|x\|_2$, $u = x/r$, and $m(u) = M(u)$, so $M(x) = r m(u)$. Let $\nabla_{\mathbb{S}} m(u)$ be the spherical gradient (the tangential part). Then*

$$\nabla M(x) = m(u)u + \nabla_{\mathbb{S}} m(u), \quad u \perp \nabla_{\mathbb{S}} m(u),$$

hence

$$\|\nabla M(x)\|_2^2 = \left(\frac{M(x)}{\|x\|_2}\right)^2 + \|\nabla_{\mathbb{S}} m(u)\|_2^2,$$

and, when $M(x) > 0$,

$$\boxed{\|\nabla M(x)\|_2 = \frac{M(x)}{\|x\|_2} \sqrt{1 + \|\nabla_{\mathbb{S}} \log m(u)\|_2^2}}.$$

Consequently the two-sided envelope $\frac{M(x_i)}{\|x_i\|_2} \leq \|\nabla M(x_i)\|_2 \leq \kappa \frac{M(x_i)}{\|x_i\|_2}$ holds at points x_i iff $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{\kappa^2 - 1}$, with $u_i = x_i/\|x_i\|_2$. Equivalently $\kappa_i = \sec \angle(\nabla M(x_i), x_i)$.

Proof. Decompose a perturbation $h = au + z$ with $a = \langle h, u \rangle$, $z = h - au \perp u$. Then $Dr(x)[h] = \langle u, h \rangle = a$ and $Du(x)[h] = (h - au)/r = z/r$ (tangent to $\mathbb{S}$). From $M(x) = r m(u)$, $DM(x)[h] = a m(u) + r Dm(u)[z/r] = a m(u) + \langle \nabla_{\mathbb{S}} m(u), z \rangle$. Setting $g = m(u)u + \nabla_{\mathbb{S}} m(u)$ and using $\langle u, z \rangle = 0$, $\langle \nabla_{\mathbb{S}} m(u), u \rangle = 0$ gives $\langle g, h \rangle = DM(x)[h]$ for all h , so $\nabla M(x) = g$. Orthogonality of the two pieces gives the Pythagorean norm identity; $m(u) = M(x)/\|x\|_2$ and $\nabla_{\mathbb{S}} \log m = \nabla_{\mathbb{S}} m/m$ give the boxed form. The envelope equivalence is immediate. $\square$

The controlling quantity. Euler fixes the radial component $m(u) = M/\|x\|$; the entire upper envelope is the *tangential* component $\|\nabla_{\mathbb{S}} \log m(u)\| = \tan \angle(\nabla M, x)$. “Can the coupling be tightened” is exactly “is the spherical log-margin gradient bounded at the support directions.” Homogeneity alone gives no such bound (it constrains only the radial part), so A is not provable from homogeneity; it requires an implicit-bias input.

4.1 Sharp linear case

Proposition 3 (Exact for linear max-margin). *For binary labels $y_i \in \{\pm 1\}$ and a linearly separable dataset, let $M_w(x) = y w^\top x$ and let $w^\star = \arg \min \|w\|_2$ s.t. $y_i w^\top x_i \geq 1$ be the hard-margin solution, with geometric margin $\gamma = \sup_{\|w\|_2 \leq 1} \min_i y_i w^\top x_i$. Then $\|w^\star\|_2 = 1/\gamma$, and at every support point ($y_i(w^\star)^\top x_i = 1$),*

$$\frac{\|\nabla_x M_{w^\star}(x_i)\|_2}{M_{w^\star}(x_i)/\|x_i\|_2} = \frac{\|x_i\|_2}{\gamma}, \quad \|\nabla_{\mathbb{S}} \log m(u_i)\|_2 = \sqrt{\left(\frac{\|x_i\|_2}{\gamma}\right)^2 - 1}.$$

So $\kappa_i = \|x_i\|_2/\gamma$ exactly: the envelope constant is the data condition number (radius-to-margin), and it is sharp.

Proof. $\nabla_x M_w = y_i w$ so $\|\nabla_x M_w\|_2 = \|w\|_2$. The norm-margin duality $\|w^\star\|_2 = 1/\gamma$ is the standard rescaling argument (normalize any feasible w to unit norm to lower-bound γ ; rescale a near-optimal unit v to feasibility to upper-bound). At a support point $M_{w^\star}(x_i) = 1$, so $\|\nabla M\|_2/(M/\|x_i\|_2) = \|w^\star\|_2\|x_i\|_2 = \|x_i\|_2/\gamma$; substitute into Proposition 2. $\square$

4.2 Conditional bound under global max-margin

Theorem 1 (Complexity-controlled max-margin envelope). *Let the model be positively homogeneous of degree $D > 0$ in its parameters, $M_{a\theta}(x_i) = a^D M_\theta(x_i)$, and let $C(\theta) \geq 0$ be a complexity with the same homogeneity, $C(a\theta) = a^D C(\theta)$, that controls the local input gradient, $\|\nabla_x M_\theta(x_i)\|_2 \leq B_i C(\theta)$ for all i . Define $\gamma_C = \sup_{C(\theta) \leq 1} \min_i M_\theta(x_i) \in (0, \infty)$ and let $\theta^\star$ solve $\min_\theta C(\theta)$ s.t. $M_\theta(x_i) \geq 1$. Then $C(\theta^\star) = 1/\gamma_C$ and, at every support point,*

$$\|\nabla_x M_{\theta^\star}(x_i)\|_2 \leq \frac{B_i}{\gamma_C} = \frac{B_i \|x_i\|_2}{\gamma_C} \cdot \frac{M_{\theta^\star}(x_i)}{\|x_i\|_2}, \quad \text{i.e. } \kappa_i \leq \frac{B_i \|x_i\|_2}{\gamma_C}.$$

If moreover $M_{\theta^\star}$ is 1-homogeneous in x , then $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq \sqrt{(B_i \|x_i\|_2/\gamma_C)^2 - 1}$.

Proof. $C(\theta^\star) = 1/\gamma_C$ is the homogeneous norm-margin duality: any feasible θ rescaled by $C(\theta)^{-1/D}$ has unit complexity and margins $\geq C(\theta)^{-1}$, so $\gamma_C \geq C(\theta)^{-1}$, giving $C(\theta) \geq 1/\gamma_C$; conversely a near-optimal unit-complexity θ_ε rescaled to feasibility has complexity $\leq 1/(\gamma_C - \varepsilon)$. Substitute $C(\theta^\star) = 1/\gamma_C$ into the gradient-control hypothesis and use $M_{\theta^\star}(x_i) = 1$ at support; the spherical form follows from Proposition 2. $\square$

Corollary 1 (Bias-free depth- D ReLU networks). *For a bias-free depth- D network with 1-Lipschitz, 1-homogeneous activations and $C(\theta) = (\|\theta\|_2/\sqrt{D})^D$ (with $\|\theta\|_2^2 = \sum_\ell \|W_\ell\|_F^2$), the input-gradient bound holds with $B_i = \sqrt{2}$: at a differentiability point with a unique rival class $j^\star$, $\nabla_x M = J_f^\top(e_y - e_{j^\star})$, so $\|\nabla_x M\|_2 \leq \sqrt{2} \|J_f\|_{2 \rightarrow 2} \leq \sqrt{2} \prod_\ell \|W_\ell\|_2 \leq \sqrt{2} \prod_\ell \|W_\ell\|_F \leq \sqrt{2} (\|\theta\|_2^2/D)^{D/2} = \sqrt{2} C(\theta)$, the last step by AM-GM. Hence, at the min-norm (global max-margin) interpolant, $\kappa_i \leq \sqrt{2} \|x_i\|_2/\gamma_{\text{net}}$.*

4.3 No universal constant (the counterexample)

Proposition 4 (The envelope constant must depend on data geometry). *There is no universal C with $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq C$ at the max-margin solution, even for linear classifiers. Fix $A \geq 1$ in $\mathbb{R}^2$ and take $(e_1, +1), (-e_1, -1), (e_2/A, +1), (-e_2/A, -1)$. The hard-margin solution is $w^* = (1, A)$ (constraints force $w_1 \geq 1, w_2 \geq A$, minimized at equality), so at $x_1 = e_1$: $M = 1, \nabla_x M = (1, A)$, radial part $(1, 0)$, tangential part $(0, A)$, hence $\|\nabla_{\mathbb{S}} \log m(e_1)\|_2 = A$ and $\kappa = \sqrt{1 + A^2} = \|x_1\|_2/\gamma \rightarrow \infty$.*

This realizes the spherical blow-up at a genuine max-margin solution; agent D (§7) exhibits the same blow-up for a bias-free degree-1 *ReLU* net ($\kappa \approx 21$ at maximal geometric margin), so the dependence on the data condition number is real for the architecture, not an artifact of linearity.

5 The lockstep, corrected (resolves B)

The conjectured inequality $\frac{d\eta}{d(\log L)} \geq c\eta$ is mis-signed for a no-go: with $d \log L / d\lambda < 0$ it only lower-bounds margin decay and permits $d \log \eta / d\lambda = 0$ (sensitivity drops at fixed margin), the opposite of an obstruction. The correct statements:

Proposition 5 (Exact lockstep on a scale orbit). *Fix a margin function $\bar{M}$ on $x_1, \dots, x_n$ and let $M_\lambda = a(\lambda)\bar{M}$ with $a(\lambda) > 0$. Then $\eta(\lambda) = a(\lambda)\bar{\eta}$, $L(\lambda) = a(\lambda)\bar{L}$, so $\eta/L \equiv \bar{\eta}/\bar{L}$ is constant and, if $a' \neq 0$, $d \log \eta / d \log L = 1$. A first-order penalty cannot change η/L by mere rescaling; it can only change it by changing the classifier’s shape.*

Proof. $M_\lambda(x_i) = a(\lambda)\bar{M}_i$ and $\nabla_x M_\lambda = a(\lambda)\nabla_x \bar{M}$ give $\eta(\lambda) = a\bar{\eta}$, $L(\lambda) = a\bar{L}$; the ratio cancels a and the log-derivatives coincide. $\square$

Proposition 6 (Implicit-function characterization; no lockstep in general). *Let θ_λ minimize $\mathcal{L}(\theta) + \lambda R(\theta)$ with $R = \mathbb{E}\|\nabla_x M\|_q$ and $H_\lambda = \nabla^2 \mathcal{L} + \lambda \nabla^2 R$. Then $\dot{\theta}_\lambda = -H_\lambda^{-1} \nabla R$, $dL/d\lambda = -\langle \nabla R, H_\lambda^{-1} \nabla R \rangle \leq 0$ (the penalty reduces sensitivity for free), and $d(\eta/L)/d\lambda = -\langle \nabla(\eta/L), H_\lambda^{-1} \nabla R \rangle$ is an alignment/operating-point quantity, not a fixed sign. In the linear case along the ridge path $w(t) = (\Sigma + tI)^{-1}b$, $d \log \eta / d \log L = m_2^2 / (m_1 m_3) \leq 1$ ($m_k = b^\top A^{-k} b$, $A = \Sigma + tI$; Cauchy-Schwarz), so η/L is non-decreasing: the penalty raises the surrogate. At a ratio-critical (robust-optimal) operating point $d(\eta/L)/d\lambda = 0$ to first order, which is where an adversarially-trained network sits; this is the honest explanation of the four-intervention experiment. The reason raising the surrogate does not raise robustness is that $\eta = \mathbb{E}[M]$ is a weak robustness surrogate (raising it can rotate the classifier away from the robust-optimal direction).*

6 The local capacity ceiling, corrected (resolves C)

As stated C is false without noise: for noiseless $x_i \sim \mathcal{N}(0, I_d)$ with $y_i = \text{sign}(u^\top x_i)$, the interpolant $g(x) = u^\top x$ has $L = 1$, $\eta = \sqrt{2/\pi}$, so $\eta/L = \Theta(1) \gg \tilde{O}(\sqrt{p/(nd)})$; a constant classifier gives $L = 0$, $\eta/L = \infty$. (This matches Bubeck–Sellke 2021, who state both that noise is necessary and that the bound needs the *supremum*, not the average, of $\|\nabla f\|$.) The salvage is a local, noisy-pair statement.

Proposition 7 (Local ceiling from close opposite-label pairs). *Let g be C^1 on a convex set, with signed margins $M_i = y_i g(x_i) > 0$ and $G_i = \|\nabla g(x_i)\|_2$. Suppose the data form m disjoint opposite-label pairs (a_s, b_s) at distances $\rho_s \leq \rho_{\max}$, each segment in the domain, with the endpoint-control*

condition $\int_0^1 \|\nabla g(\gamma_s(t))\|_2 dt \leq \frac{C}{2}(G_{a_s} + G_{b_s})$ for some $C \geq 1$. With $\eta = \frac{1}{2m} \sum_s (M_{a_s} + M_{b_s})$, $L = \frac{1}{2m} \sum_s (G_{a_s} + G_{b_s})$, then $\eta \leq \frac{C\rho_{\max}}{2}L$, i.e. $\eta/L \leq C\rho_{\max}/2$.

Proof. For a pair, $M_a + M_b = g(x_a) - g(x_b) = \int_0^1 \langle \nabla g(\gamma(t)), x_a - x_b \rangle dt \leq \rho \int_0^1 \|\nabla g(\gamma(t))\|_2 dt \leq \frac{C\rho}{2}(G_a + G_b)$ by FTC, Cauchy–Schwarz, and endpoint control; sum over pairs and divide by $2m$. $\square$

This is the honest local analogue of the law of robustness: a ceiling needs close opposite-label forcing plus no gradient spikes on the connecting segment.

7 The bracket condition number (resolves D)

The certificate looseness is $\kappa_{\text{br}} = Lr_2/\eta \geq 1$, with $\kappa_{\text{br}} = 1$ iff M is affine on the segment from x to the boundary, so κ_{br} measures failure of local linearity, a curvature quantity, not a margin quantity. D as stated (a bound $g(\mu)$ in the separation margin alone) is false: a bias-free degree-1 ReLU net can have $\kappa_{\text{br}} \approx 21$ at *maximal* geometric margin. Two true statements:

Proposition 8 (Architectural bound; benign curvature). (P1) *For any bias-free degree-1 homogeneous net, $\kappa_{\text{br}} \leq \sec \angle(\nabla_x M(x), x)$, the same envelope constant as Proposition 2; in particular the polynomial $t \mapsto t^{2n+1}$ (degree > 1) blow-up is excluded ($\kappa = 2n + 1$ there comes from non-unit homogeneity).* (P2) *If M has curvature β along the path with $2\beta\eta \leq L^2$ then $\kappa_{\text{br}} \leq 2/(1 + \sqrt{1 - 2\beta\eta/L^2}) \leq 2$ (a CURE / local-linearity-regularization style bound).*

P1 ties the bracket looseness to the *same* spherical/angular quantity as the envelope, so η/L is loose exactly when the input gradient is tangentially misaligned with the datum; the implicit bias does not certify the benign regime (min-norm interpolants permit localized curvature spikes; this is why curvature/local-linearity regularizers exist).

8 The one controlling quantity, and what remains open

Synthesis. A single quantity governs both the envelope and the certificate: the spherical log-margin gradient $\|\nabla_{\mathbb{S}} \log m(u_i)\| = \tan \angle(\nabla M, x_i)$ at support directions, with $\sec \angle = \sqrt{1 + \tan^2}$ equal to the envelope κ (Prop. 2) and an upper bound on the bracket κ_{br} (Prop. 8). It is exactly the data condition number $\|x\|/\gamma$ for linear max-margin (Prop. 3), bounded by $B_i\|x\|/\gamma_C$ under global max-margin with gradient-controlling complexity (Thm. 1, Cor. 1), and unbounded across datasets (Prop. 4). The margin alone never bounds it.

The open path. Theorem 1 and Corollary 1 assume the trained network is the *global* max-margin (min-complexity) interpolant. Gradient descent on homogeneous networks converges only to a *KKT point* of the max-margin program (Lyu–Li 2020; Ji–Telgarsky 2020), and such a point need not be a local or global max-margin optimum for ReLU networks (Vardi–Shamir–Srebro 2022, arXiv:2110.02732). So the missing theorem is:

Conjecture 1 (Implicit-bias spherical-gradient bound). For the gradient-descent KKT solution on a bias-free homogeneous network with data of condition number $\|x\|/\gamma$, the spherical log-margin gradient at support directions satisfies $\|\nabla_{\mathbb{S}} \log m(u_i)\|_2 \leq g(\|x\|/\gamma)$ for an explicit g (e.g. $g = \sqrt{(\|x\|/\gamma)^2 - 1}$ as in the global case), absent the global-max-margin assumption.

This is the genuinely open piece. Candidate routes: (i) strengthen the KKT structure of Lyu–Li/Ji–Telgarsky to control per-neuron coherence at support points; (ii) a curvature/local-linearity bound on the GD trajectory (CURE/LLR-style) that discharges the benign regime of Proposition 8; (iii) accept the global-max-margin hypothesis and state the result conditionally (the honest current option, used in the paper). Open residuals: B in the genuine non-scale, non-ridge nonlinear regime; C in the noisy, $p \ll nd$, training-point-average niche (needs an average-not-sup argument); D’s explicit GD dataset realizing large κ_{br} at large margin.

Statement	Status	Where
Radial (Euler) lower envelope	proved (standard)	Prop. 1
Ratio-maximization degeneracy	proved (known)	Lem. 1
Exact radial–tangential split	proved (standard polar)	Prop. 2
Linear max-margin: $\kappa = \ x\ /\gamma$	proved, sharp	Prop. 3
Conditional envelope (global max-margin)	proved (adapted duality)	Thm. 1, Cor. 1
No universal constant	proved (counterexample)	Prop. 4
Scale-orbit lockstep / IFT characterization	proved	Prop. 5, 6
Local noisy-pair ceiling	proved	Prop. 7
Bracket $\kappa_{\text{br}} \leq \sec \angle$; benign ≤ 2	proved	Prop. 8
Implicit-bias spherical-gradient bound	<i>open</i>	Conj. 1

Provenance (for honest naming later). Prop. 1, Lem. 1, Prop. 2 are standard/known (Euler identity; Tsuzuku; polar gradient decomposition). Prop. 3 and Thm. 1 are adaptations of the norm–margin duality. Cor. 1 is a direct application of the product-of-spectral-norms Lipschitz bound. Prop. 4 (no universal constant) and the identification of the spherical log-gradient as the controlling quantity are the genuinely informative observations; Conj. 1 is the open contribution target.